Seneca POLYTECHNIC

Final Case Project

“MOVVO: Marketing Location Based Data”

Group 1

BAN 120 - Analytics Case Studies

Rita Obaji

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Kevin Lu

Wei Jhen Jhang

Omar Nolasco Montesinos

Saurabh Singh

Aathithyan Sivalingam

Min Zhang

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BAN120 – Template for Case Analysis

IDENTIFICATION OF PROBLEM(S) **(Kevin)**	Movvo face too many potential markets and needs to choose one market that is the right fit for this company. However, Moovo is unsure which market would be most compatible with customer needs, leading to lack of clarity. Another problem is internal conflict since the founders disagree on target market - Roberto wants retailers, Suzy wants malls, and Diana wants transportation hubs. However, entering multiple markets rises competition and creates challenges in differentiation. Entering more markets likely to get expose Movvo to strong competition, increasing pressure and requiring clearer communication of its superiority. Another issues is privacy concerns, which pose risk for retailers and may create negative brand perceptions affecting customers and the market. For example, a pilot project was cancelled due to concerns of privacy on retailer marketing department. Movvo must demonstrate its value by clearly explaining the difference between "big data" and "increased wisdom" and proving it offers real business value. Movvo's pricing and business model appear uncertain, as issues like mismatched tags or shrinkage could affect cash flow and customer service. (Hussain, August 14, 2025) If Movvo targets airports, the market would be extremely expensive and slow-paced, delaying its ability to reach customers. Internal debates may arise, with some wanting to focus on one industry to avoid low ambition, while others prefer multiple industries despite limited resources. Lastly, Movvo must convince investors by having a clear market focus, a monetization plan, and a realistic growth strategy.
DECISION CRITERIA (Aathithyan)	● Understanding which targeting customers benefit from its technology (Retailers, Malls, Transport and Mobility, and Airports) ● Identifying which target market has the largest revenue potential ● Being able to utilise its technology to stand out against its competitors ● Avoiding legal concerns from Privacy risks ● Ensuring Movvo can convince its potential customers with its demonstration strategy

	● Finding the right business model that would suit their business goals in scalability and growth ● Planning on how to stay competitive and innovative with its existing competitors ● Providing a low-cost solution for the targeted customers

ANALYSIS

(Omar)

Movvo is trying to decide between four possible markets, to make the decision, two different aspects will be evaluated:

1. Projected Revenue

We will use a "proxy" to know the approximately cost per location based on the pricing of big consulting firms:

Location Size	Assumed Complexity / Features / Effort	Suggested Annu Fee per Locati
Small (small store / boutique / small outlet)	Basic beacon setup, simple analytics	**USD 10,000**
Medium (mid-size store, small mall wing, regional outlet)	More beacons, richer analytics + modest support	**USD 15,000**
Large (large mall, big retail store, mid-size airport terminal)	Full sensor deployment, advanced analytics + regular maintenance & support	**USD 30,000**
Huge (major mall, large airport, multi-floor shopping complex, transit hub)	High complexity: many beacons, heavy data volume, advanced analytics, custom dashboards, support + dedicated account management, possibly integration with other systems	**USD 50,000**

Using that pricing we can forecast the revenue:

Rank	Market	Locations in Case	Fee per Location	Total Possible Market Revenue	% of Reach	Projected Annual Revenue
1	Retailers	250,000 stores	$10,000	$2.50B	5%	$125M
2	Shopping Malls	12,000 malls (US 6,106 + EU 5,721)	$30,000	$354.8M	10% US + 5% EU	$26.9M
3	Airports	45 airports (>10M passengers)	$350,000	$2.25M	75%	$18M
4	Transport Agencies	7,000 agencies	$15,000	$105M	15%	$16M

2. Alignment Between Movvo's Core Advantages and Each Market Segment

Movvo Advantage / Customer Pain Point	Retailers	Shopping Malls	Transport Agencies	Airports
Need for High-Precision Location	Moderate, impacts are difficult to measure	High, flow is important for pricing	Moderate, Passanger flow	Critical, crowd contr security
Urgency of Problem (Rapid installation)	Low–Medium (optional analytics)	Medium (traffic counts already exist)	Medium (and slow to adopt due to regulation)	Very High (Congesti security queues, dela
Privacy as a Critical Competitive Differentiator	Low–Medium	High	Medium	Critical
Rank	4	2	3	1
Result	Low alignment with Movvo's strengths	Almost aligned with Movvo's strengths	Moderately aligned with Movvo's	**Perfectly aligned w strengths**

Using the ranks of both perspectives, the best options for Movvo are Shopping Malls and Airports. To make the final decision, we will focus

on the margin levels between the options. This margin will be a possible investment for Movvo:

- Airports have an 81% margin and usually have dedicated budgets for innovation and safety.
- Malls have constrained budgets, heavily dependent on tenant rent and foot traffic revenue. Innovation spending competes with marketing campaigns, renovations, and retail incentives with no dedicated budget.

ALTERNATIVES

(Viola)

Alternative 1: Mall Focus

Target: Large Malls in the US and Europe

Positioning: maximize their rental income, optimize the store effectiveness of in-mall marketing efforts and shopper experience

Business Model: Software/Hardware Provider. Customers pay full costs upfront.

Key Analysis Points:

- Market: High customer acceptance, low sales cost.
- Advantage: Lowest funding risk for Movvo.
- Challenges: 1.Data exhaustion issue needs to be addressed. 2.Resistance due to the upfront payment requirement may slow adoption.

Alternative 2: Retail Focus

Target: Large Apparel or FMCG Retailers (Global Top 250).

Positioning: Analytics Engine for Physical Stores. Use Movvo's technology to optimize product assortment.

Business Model: Consulting-Based Model. Free installation; profit comes from high valued, evidence-based consulting services.

Key Analysis Points:

- Market: Largest size, but highest sales cost and lowest success rate.
- Advantage: higher acceptance as Movvo install its software and hardware at a low cost

- Challenge: Requires funding for hiring and training consultants, and it is not the strength of Movvo.

Alternative 3: Transport Focus

Target: Large Mass Transportation companies.

Positioning: soft security & crowd control systems. Improve simplicity and user-friendliness of its services

Business Model: different pricing model with commission. Low entry fee + 5% commission on ticket sales.

Key Analysis Points:

- Market: Large recurring potential ($120 million a year in U.S. and Europe).
- Advantage: Revenue is directly tied to customer usage/success.
- Challenge: High complexity due to integration with public and financial systems. Movvo has to define how to demonstrate the value and overcome customers' resistance to adoption and the best pricing and scaling model.

DECISION(S)/ RECOMMENDATIONS (Chalres)

Movvo must focus its initial market strategy on Airports and adopt a Subscription-as-a-Service (SaaS) business model. The airport segment is the most attractive, offering the highest projected gross profit margin of 81% and a strong expected conversion rate of 75%. Airports have critical needs related to crowd management and passenger monetization, which directly benefit from Movvo's superior 10-fold location accuracy. Furthermore, the SaaS model is recommended because it eliminates the initial Capital Expenditure (CAPEX) for customers, easing adoption and overcoming the resistance caused by "data exhaustion" and large upfront investment. This recurring revenue model is also highly favored by investors.

To successfully enter this market, Movvo must position its offering as the most accurate, low-cost location intelligence solution, highlighting its unmatched technical precision. Implementation requires actively promoting the technology's guarantee of completely anonymized data collection to mitigate the privacy concerns that severely affected competitors. Finally, to educate prospective customers and demonstrate

	quantifiable value, Movvo should continue its efforts to collaborate with academic researchers to produce external, evidence-based knowledge based on Movvo's data.
IMPLEMENTATION (Saurabh)	In order to successfully implement the recommended strategy, Movvo should focus on airports as the primary entry point for the business. As with any business model, it is advisable to start slowly by launching a pilot program targeting a smaller airport with around 10 million annual passengers. For handling such a large dataset, the SaaS model works effectively, and the pilot program would generate evidence of improved crowd flow, reduce delays, and increase commercial revenue inside the terminal. For revenue generation, Movvo can introduce a subscription-based pricing model by offering airports an annual fee that includes the analytics dashboard, system maintenance, and ongoing support. This aligns well with airport operators' preference for operational (OPEX) expenses rather than large upfront investments. Movvo must also focus on the privacy aspect to ensure that all tracking remains fully anonymous. The company should prepare clear communication materials that airports can use with travellers and stakeholders to address potential concerns. Finally, Movvo should invest in research and academic partnerships to analyse pilot program data and develop optimal strategies for implementing the solution at a larger scale.

Questions + Answer

Answer all the questions below on a separate page:

1. Given the facts of the case, explain how Movvo effectively communicate the difference between "big data" and "increased wisdom" to prospective clients.

The "big data" is the large amounts of raw information and movement patterns. Typically, competitors entering the market provide counts, traffic volumes, or location pings without the need of deep analysis. Without proper guidance and support, this data can become overwhelming and cause confusion. With big data, the focus is on quantity-more stock numbers, more signals, and more tacking events. Competitors already have a lot of data, but Moovo may face difficulties turning it into meaningful decisions.

The "increased wisdom" means actionable insights that help customers improve decisions, layouts, crowd flow, and revenue. For Movvo, this means using real use cases to manage crowded areas, optimize product placement, and identify bottlenecks. Movvo could provide visual dashboards so clients can understand behavioral patterns rather than just raw numbers. This results in evidence-based outcomes instead of simple data streams, supported by pilot studies and academic research. Therefore, Movvo positioned itself as a provider of interpreted knowledge instead of a data collection tool. "Increased wisdom" focuses more on quality-smarter decisions, operational improvements, and measurable ROI.

2. Describe the advantages and disadvantages of Movvo targeting shopping malls versus transportation hubs for its initial market entry strategy.

The advantages of Movvo targeting shopping malls where shopping malls know the volume of foot traffic well which is less resistance to new tracking tech, great price for sale cost per mall, malls have exist budget for traffic analysis to see how busy they are and retail optimization to find out how was their profits going and ensures meeting the target. Able to have clear financial value where Movvo helps with the mall to optimize store mix, marketing activities, rental pricing, and offering clear financial value. There could have market size because of large number of malls in US and Europe. The disadvantages of Movvo targeting shopping malls are very competitive as another retailers store already sell with similar products, quantities, and solutions. Market can be fragmented as Movvo must approach many individual mall owners. Lastly, mall decision may be applied to privacy/public perception concerns.

The advantages of Movvo targeting transportation hubs which able to have crowd monitoring and flow optimization which can lead to higher accuracy. Movvo get an offer for recurring revenue model. Movvo could get biggest market size and more transit companies. Movvo would be able to solve real organizational plan like volume of passengers, counting billing, congestion

management, and complaints. The disadvantage Movvo targeting transportation hubs are more fragmented market as many of agencies tend to sell to, highly regulated due to slow and bureaucratic procurrenemt processes. It must have higher expectations and technical complexity as it is needed for extremely reliable system. It can increase adoption risk if unions, operators, or governments resist changes in passenger tracking. Lastly, there would be integration with ticket/payment systems could add more complexity.

3. Summarize how Movvo's technology outperform competitors like Euclid, Nomi, and Experian Footfall, and explain how this should be reflected in its marketing Strategy?

Movvo is able to outperform its competitors because it uses mobile phones to find and track a person's location. It can track not only indoors but also in outdoor spaces, which makes it unique from its existing competitors that are focused mainly on indoor location solutions (Camacho et al.7). Movvo is able to provide a strong location-based data solution by detecting radio frequency signals through a hardware system that can be installed in any indoor environment. This hardware system provides real-time data with high accuracy using beacon technology. Beacon technology works by detecting antenna patterns that react to signals from customer mobile phones (Camacho et al.8). The hardware setup includes multiple antennas that respond to the beacon signals, allowing Movvo to produce accurate and cost-saving retail data over time. Movvo's beacons can cover up to a 1km radius, and the location accuracy is based on the distance between the mobile device and the nearest beacon station that is placed indoors (Camacho et al.8). It also stands out from its competitors as Movvo's tracking data remains anonymous and is purely based on mobile radio signals, giving retailers real-time insights without identifying the specific user.

Movvo's technology strengths should be implemented in its marketing strategy by highlighting the benefits that differentiates the company from its competitors. In this case, Movvo's four main benefits are Location Accuracy, Low-Cost Solution, Anonymised Technology, and Privacy Guarantees (Camacho et al.8). For Location Accuracy, Movvo was able to achieve a 10-fold improvement in its location precision (Camacho et al.8). Movvo should mention this benefit since retailers can use this location accuracy to identify high-density consumer zones, so retailers can place their products in more high-density consumer locations. This leads to a higher chance of sales due to the amount of potential customer presence. Movvo should also mention its low-cost hardware solution as a major advantage compared to its competitors, as they were able to deliver a strong performance while having a low cost in their technology compared to its competitors.

Lastly, Movvo should mention that their technology is anonymised and supports privacy guarantees. Their system avoids identifying individuals and even gives customers an option to choose whether they want to be monitored from their mobile (Camacho et al.9). Implementing this feature can limit the full potential of the technology, but in the long term, it helps the

company avoid any legal privacy concerns. This opportunity should be utilised in order to promote its business. Overall, by highlighting its benefits, Movvo can strengthen its marketing strategy and build trust with retailers and potential consumers who want to use its technology in order to scale its business to further success.

4. Illustrate the strategies which Movvo adopts to address privacy concerns that are raised by retailers and consumers, particularly considering the failed pilot in the Netherlands?

Movvo understood that privacy can be a critical challenge. The company has already experienced this risk when a previous pilot in the Netherlands failed because the marketing team worried it could damage the brand.
To solve this, Movvo focused on a fully technical solution: the system was built to collect data that was completely anonymized. It captures real-time movement of mobile devices without individual identification. Movvo believed this would position them as a more privacy-enhanced alternative to invasive tools like cameras.
When it comes to commercial use, Movvo planned to give customers two options: adopt an opt-in system for full transparency, or rely on Movvo's anonymous tracking model and communicate why it doesn't create privacy concerns.
Finally, to build credibility and show that the data has value, Movvo began developing partnerships with academic researchers. By allowing them to access the data for free, Movvo hoped external validation would help reinforce the legitimacy and benefits of their technology.

5. Describe how Movvo's trilateration technology using beacon signals provides superior location accuracy compared to GPS and RFID-based systems.

Movvo takes advantage of the constant signalling that smartphones perform when they search for Wi-Fi and Bluetooth connections. This signalling is picked up by strategically placed beacons, usually in a "triangulated way" (hence the name trilateration), which send the information to a HUB to translate it using its own developed algorithm, converting it into highly accurate, low-cost real-time location estimates. The estimates are enhanced using existing antenna networks inside buildings, resulting in an advantageous solution for indoor location and consumer behaviour in physical stores.

Compared to other technologies Movvo is position as follows:

Technology	How It Works	Accuracy Level	Why Movvo Is Superior
GPS	Uses satellites, location is calculated based on distance to satellites in orbit.	**approx 5 meters outdoors**, often **unusable indoors**.	Movvo uses **local indoor beacons**, so signals are stronger and more reliable, enabling **much higher precision** than GPS inside buildings (10 times better).
RFID-Based Systems (shopping cart tagging)	RFID tags attached to shopping carts or objects are detected by fixed and expensive readers.	Low to moderate precision (coarse zones, not exact positions).	Movvo provides **continuous exact coordinates** instead of broad detection zones. Higher accuracy and no tagging or special equipment required.

6. Explain the aspects of Movvo's system architecture that contributes to its lowcost implementation, and describe how might this influence its adoption rates? (Charles)

Movvo's system architecture facilitates a low-cost implementation primarily by strategically minimizing the necessity for clients to invest heavily in new equipment, thereby driving adoption. The core cost advantage stems from Movvo's ability to leverage existing assets and infrastructure already present at client locations, such as the network base stations and beacons utilized by mobile operators for seamless data flow. This dramatically reduces the client's upfront Capital Expenditure (CAPEX), as the solution is engineered to use standard, off-the-shelf technologies and beacons. Clients only need to install a few inexpensive antennas (beacons) to increase signal granularity and a simple desktop computer to receive, process, and store the signal. The company's proprietary algorithm then inexpensively improves the signal information using trilateration, contributing further to the system's overall cost-efficiency.

This low-cost architecture is a powerful lever for influencing adoption rates. By substantially reducing the initial investment required, Movvo helps overcome customer resistance and the concern of "data exhaustion," where companies are reluctant to commit a large budget to yet another new big data solution. Crucially, the system's low cost makes the highly attractive subscription-based business model feasible. Movvo can absorb the minimal installation costs, eliminating the barrier of client CAPEX, which is expected to make it easier to convince customers to adopt the technology. Ultimately, the combination of its low cost and its superior location accuracy creates a compelling Unique Selling Proposition (USP) for Movvo.

7. Summarize the pricing and business model options which Movvo can use to attract both investors and customers while maintaining scalability. State how Movvo can balance the pressure from investors to focus its market strategy with the founders' differing views on target sectors? (Saurabh)

Movvo primarily has three revenue-generating business models. The first option is to sell hardware and charge an upfront cost for providing the software solution. The second, and more effective option, is to offer a subscription-based SaaS model. The third option involves providing consulting services based on their location insights. The SaaS subscription model is the most suitable choice because it removes the large upfront cost that often prevents customers from adopting new data technology. With this approach, Movvo can charge an annual fee along with a small installation cost, making it easier for airports to try the system while still creating predictable recurring revenue for investors.

To manage investor pressure, Movvo should focus on airports as the primary market. Airports allow Movvo to demonstrate the true value of its business model, as they have a high need for crowd management, strong demand for accurate analytics, and lower price sensitivity. Once Movvo becomes strong and successful in the airport sector, it will naturally increase their chances of expanding into other sectors like retail and transportation. This approach provides both focus and long-term growth potential, satisfying investors as well as the founding team.

8. Illustrate the concepts of the case difficulty cube and explain how you would position the case on the cube. List and explain the major concepts in the case. (Saurabh)

The case difficulty cube explains how challenging a case is based on several factors, but most importantly it depends on three things: how much analysis is required, how many concepts need to be understood, and how complicated the overall situation is.

The Movvo case is considered high in difficulty because it compares several markets such as retail, transportation, and the aviation sector, while also evaluating different pricing models. It involves both technical and business aspects, including ideas like privacy, big data, and choosing the right target customers. This makes it medium to high in conceptual difficulty.

The Movvo case is also complex because it deals with different industries, strong competitors, and investor pressure, while the founders themselves disagree on the company's direction. Due to these factors, the Movvo case fits in the top-right area of the difficulty cube, meaning it has high analytical and high contextual difficulty.

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